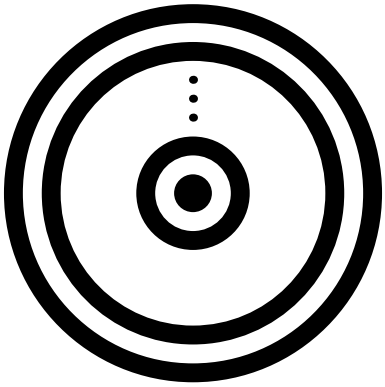
2023 WSMO Team Round

SMO Team

TR 1: Bob has a number of pencils. When Bob splits them into groups of 10, he has 3 left over. When he splits them into groups of 12, he has 5 left over. Find the smallest number of pencils Bob can have.

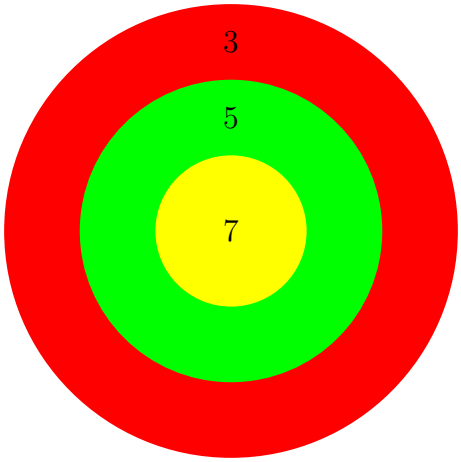
TR 2: Positive integers W, S, M, and O satisfy $W + S + M + O = 13$. Find the maximum possible value of $WSMO$.

TR 3: In the figure below, there are 1023 total circles with radii $1, 2, \ldots, 1023$. The area between circles alternates between shaded and non-shaded. The area of the shaded region can be expressed as $k\pi$. Find the remainder when k is divided by 1000.



TR 4: Honko the hamster is in his cage. He wants to find the smallest distance needed to travel to reach four tennis balls. His current position is $(0, 0)$. The tennis balls are located at $(1, 1)$, $(2, -2)$, $(-3, -3)$, and $(-4, 4)$. The length of the shortest path can be expressed as $\sqrt{a_1} + \sqrt{a_2} + \cdots + \sqrt{a_n}$, where $a_1, \ldots, a_n$ are positive integers and n is as small as possible. Find $a_1 + a_2 + \cdots + a_n$.

TR 5: A monkey is throwing darts at the dart board pictured below. The dart is equally likely to land anywhere on the board. Point values for the three regions are labeled and the radii of the three circles are 1, 2, and 3, respectively. The expected value of points the monkey gets from 5 dart throws can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.



TR 6: A quartic real polynomial $f(x)$ satisfying $f(3+2i)=0$ has 3 distinct roots that sum to 12. Find the product of all the roots of f.

TR 7: In triangle ABC, $AB=13$, $AC=14$, and $BC=15$. A rectangle R is inscribed such that one side of R lies along a side of ABC and the other two vertices lie on the remaining two sides. Suppose the area of R is maximized. The minimum possible ratio of the length of the side of R lying along a side of ABC to its adjacent side length can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

TR 8: Let $f(x)=\sqrt{x-\sqrt{x-\sqrt{x-\ldots}}}$. Find the remainder when the least integer a such that $f(f(f(f(f(a)))))$ is a positive integer is divided by 1000.

TR 9: A circle ω has three chords of equal length, $4+2\sqrt{3}$, which intersect forming a triangle with side lengths 2, 2, and $2\sqrt{3}$. The square of the radius of ω can be expressed as $a-b\sqrt{c}$, where a, b, and c are positive integers and c is not divisible by the square of any prime. Find $a+b+c$.

TR 10: Let square $ABCD$ be a square with side length 4. Define ellipse ω as the ellipse that is able to be inscribed inside $ABCD$ such that 2 of its vertices on its minor axis and 1 of its vertices on its major axis form an equilateral triangle. The largest possible area of ω can be expressed as $m\pi\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m+n$.

TR 11: When $\frac{1}{7}$ is expressed in base k, the digits are $0.a_{k,1}a_{k,2}\ldots$, where $a_{k,1}, a_{k,2}, \ldots$ are base-k digits. Let $f(p)$ denote the minimum positive integer $x \geq 2$ such that $a_{p,1}=a_{p,x}$, for $p \not\equiv 0 \pmod 7$, and 1 for $p \equiv 0 \pmod 7$. Find $\sum_{i=2}^{100} f(i)$.

TR 12: Consider parabola $\mathcal{P}$ pointing upwards with vertex at the origin of the Cartesian plane. Denote the focus of it as F and the directrix of it as $\mathcal{L}$. Point P with x-coordinate 4 is selected on $\mathcal{P}$. The perpendicular bisector of FP meets $\mathcal{L}$ at Q. Given the x-coordinate of Q is 3, then FP^2 can be expressed as $\frac{a+b\sqrt{c}}{d}$, where a, b, c, and d are positive integers, c is not divisible by the square of any prime, and $\gcd(a, b, d) = 1$. Find $a + b + c + d$.

TR 13: Gerry the Squirrel is at a corner of a 9×9 grid of points. An acorn is placed randomly on a different point of the grid. Four coins are placed randomly on the grid, none of which are located on Gerry's starting point or on the acorn. Gerry will take a shortest path to the acorn, and if there are multiple shortest paths, Gerry will pick one randomly. The expected number of coins Gerry passes during his trip can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

TR 14: Consider quadrilateral $ABCD$ with circumcircle centered at O. Given $AB = 2 < CD$, the circumcircle of $\triangle ABO$ is tangent to the circumcircle of $\triangle COD$. The circumcircle of $\triangle AOD$ passes through the center of the circumcircle of $\triangle ABO$. The radius of the circumcircle of $ABCD$ is 6. Let M and N be the area and perimeter of $ABCD$, respectively. The value of $\frac{M}{N}$ can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a + b + c$.

TR 15: On a number line labeled 0, 1, 2, 3, 4, 5, an old man starts at 0 and tries to reach 5. Initially, he knows to walk right. However, he has dementia. On each move, there is a $\frac{1}{3}$ chance he forgets which direction he is supposed to go, resulting in him walking the opposite direction. If the old man tries to walk left when he is at 0, he falls off a cliff and dies. The probability the old man reaches 5 without dying can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.